

Outer X-matrix block-encoding comparison: up vs down

Parameters: $N_{\text{up}} = 4$, $N_{\text{down}} = 22$, N_{IP} (central bond) = 256
 $N_k = k^3$ for k in $[1, 2, 3, 4, 5, 6]$ (so N_k in $[1, 8, 27, 64, 125, 216]$)
phase_bitsize $b = 32$

This isolates the OUTER block encoders of the interpolating-vector matrices X only -- the central Coulomb kernel W is excluded. Both direct and exchange use the same outer encoders, so these costs are shared across all four schemes. Each is extracted from the full BE (ExchangeCoulombBlockEncoding.B_up / .B_down, identical to the direct BE's B_mu / B_lam).

X_{up} : $\sum_k |k\rangle\langle k| (x) X^{\{\text{up},k\}}$ ($N_{\text{up}} \times N_{\text{IP}}$ isometry via ReflectionRectangularBlockEncoding, $n_{\text{reflections}} = \min(N_{\text{up}}, N_{\text{IP}}) = 4$).
 X_{down} : $\sum_k |k\rangle\langle k| (x) X^{\{\text{down},k\}}$ ($N_{\text{down}} \times N_{\text{IP}}$ isometry, $n_{\text{reflections}} = \min(N_{\text{down}}, N_{\text{IP}}) = 22$).

Both isometries are padded to $n_{\text{rows}} = \text{next_pow2}(\max(N, N_{\text{IP}})) = 256$, so the cost scales with the reflection count (= short dimension), not the padded size -- hence X_{down} (22 reflections) costs $\sim 5.5 \times X_{\text{up}}$ (4 reflections).

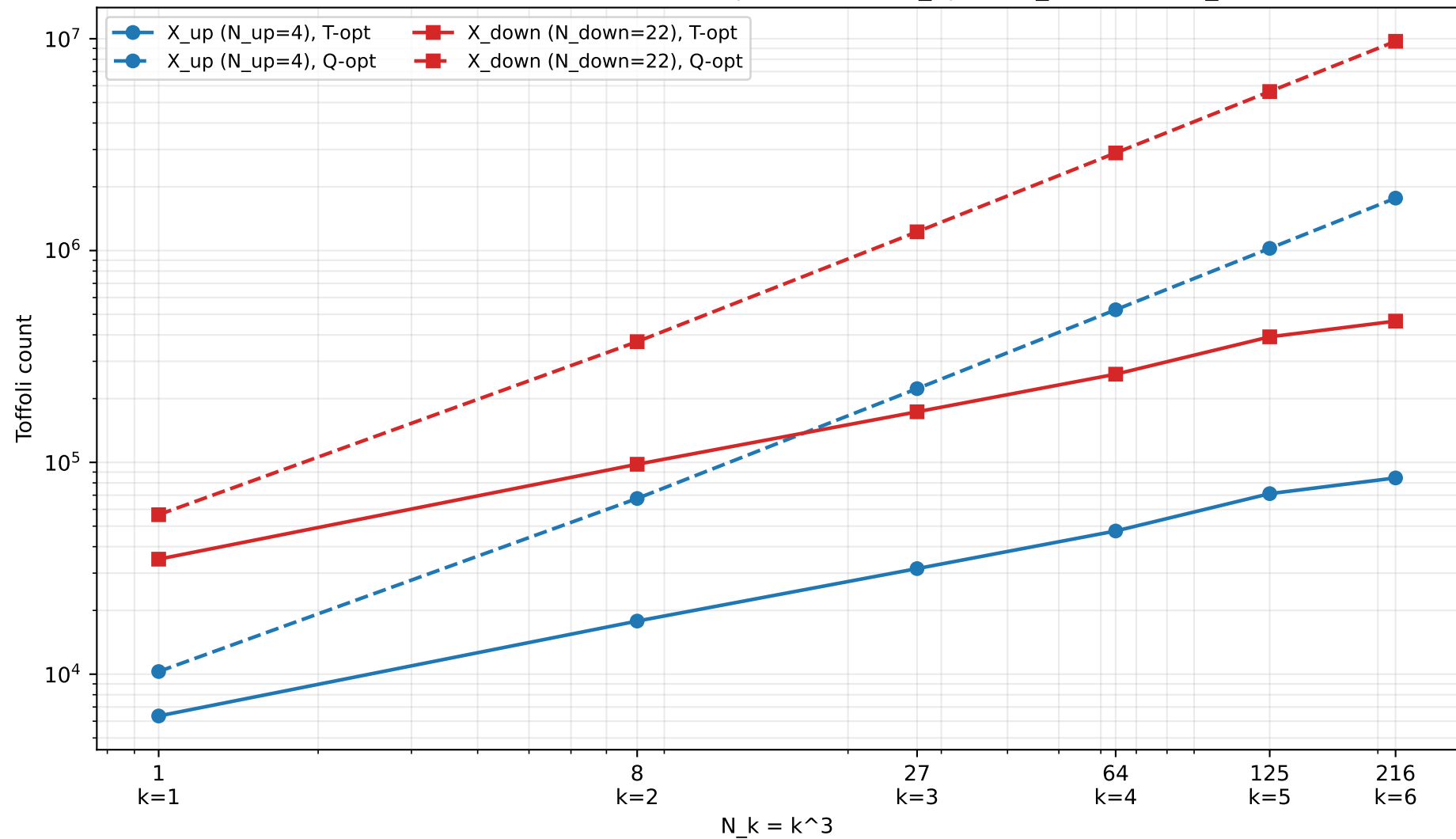
Tuning policies (per curve):

T-opt: every QROAM log_block_sizes at the closed-form Toffoli optimum.

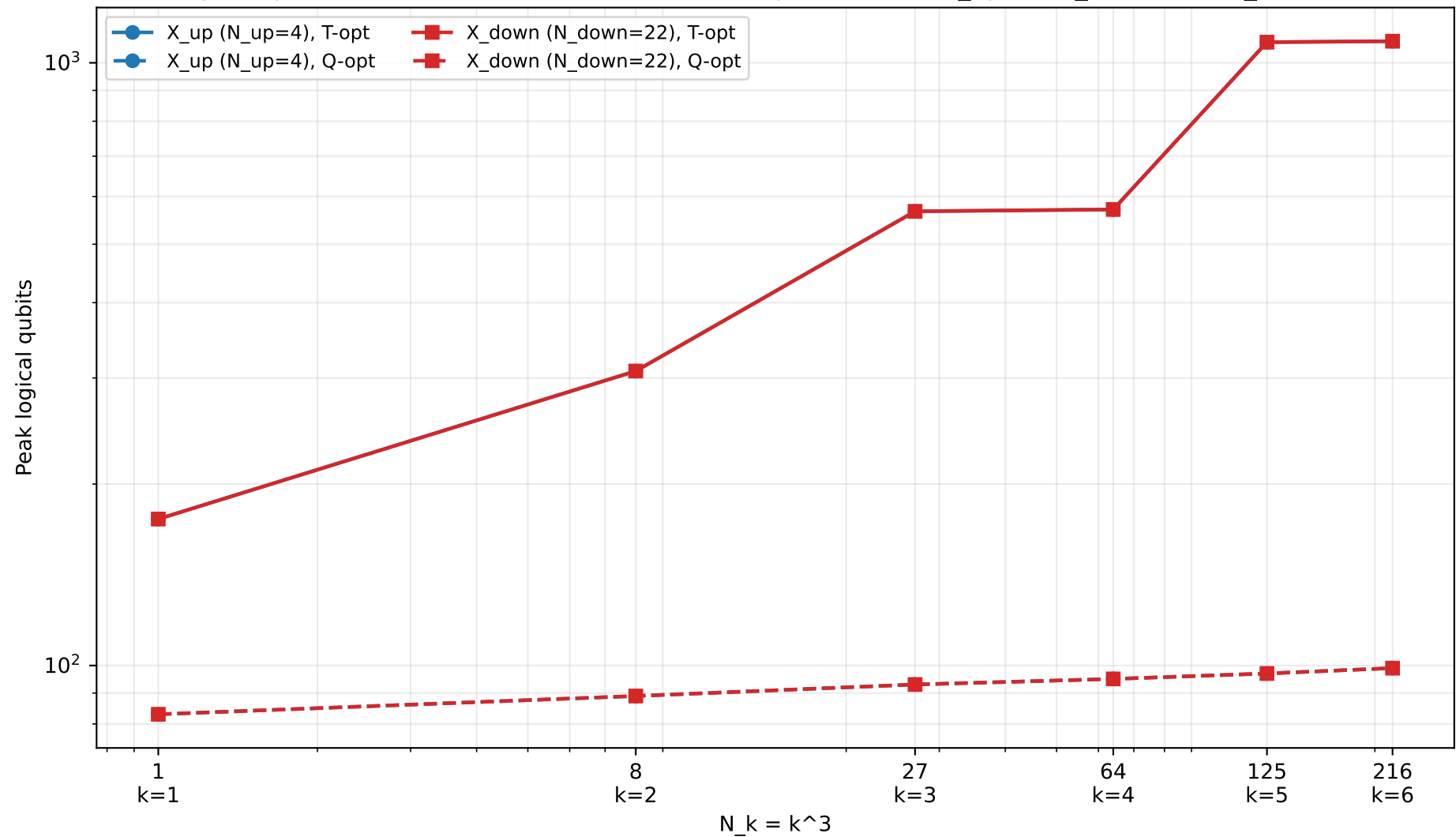
Q-opt: no QROAM blocking (log_block_sizes = 0) -> minimal qubits
(qubit count then grows only logarithmically in N_k).

All Toffoli / qubit counts come from Qualtran's QECGatesCost / QubitCount.

Toffoli count: outer X-matrix block encoders (up vs down) ($N_{\text{up}}=4$, $N_{\text{down}}=22$, $N_{\text{IP}}=256$, $b=32$)



Peak logical qubits: outer X-matrix block encoders (up vs down) ($N_{\text{up}}=4$, $N_{\text{down}}=22$, $N_{\text{IP}}=256$, $b=32$)



Toffoli count: outer X-matrix block encoders

X encoder	N_k=1	N_k=8	N_k=27	N_k=64	N_k=125	N_k=216
X_up (N_up=4), T-op	6,348	17,812	31,508	47,420	71,172	84,436
X_up (N_up=4), Q-o	10,292	67,524	222,868	525,380	1,024,116	1,768,132
X_down (N_down=2	34,914	97,966	173,294	260,810	391,446	464,398
X_down (N_down=2	56,606	371,382	1,225,774	2,889,590	5,632,638	9,724,726

Peak logical qubits: outer X-matrix block encoders

X encoder	N_k=1	N_k=8	N_k=27	N_k=64	N_k=125	N_k=216
X_up (N_up=4), T-op	175	308	567	571	1,082	1,086
X_up (N_up=4), Q-o	83	89	93	95	97	99
X_down (N_down=2	175	308	567	571	1,082	1,086
X_down (N_down=2	83	89	93	95	97	99